



Unveiling the Media Landscape: A Data Analysis of 2024 U.S. Election Coverage

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The Objective + Data



Objective

- Personal experiences of growing up during polarizing times
- Concerns about misinformation on platforms like Facebook and TikTok
- **Question: "Are American news outlets using titles and content that propagate polarization amongst the two parties for the 2024 presidential election?"**

Data + Database

- Mongo as the NoSQL database → JSON, aggregation functionality
- Utilization of **News API** that scrapes news article information
- Free plan limited data to the past month



News API Functionality

- Imported package into Python for API calls
- Filtered on our chosen keywords
- Example of News API gathering articles based on a keyword

```
← All articles about Tesla from the last month, sorted by recent first →

GET https://newsapi.org/v2/everything?q=tesla&from=2024-03-09&sortBy=publishedAt&apiKey=API_KEY

{
  status: "ok",
  totalResults: 7338,
  - articles: [
    - {
      - source: {
        id: null,
        name: "Biztoc.com"
      },
      author: "aol.com",
      title: "Anything but Tesla: US shoppers want more EV options",
      description: "None EV shoppers want more choices than Tesla. • None This is the latest sign that the EV slowdown is catching up to Elon Musk. Electric vehicle shoppers are getting sick of Tesla. Well over half of the EV shoppers surveyed by UBS in an annual study said they...",
      url: https://biztoc.com/x/86f38b7826bf7ce8,
      urlToImage: https://c.biztoc.com/p/86f38b7826bf7ce8/s.webp,
      publishedAt: "2024-04-09T18:58:12Z",
      content: "None EV shoppers want more choices than Tesla.None This is the latest sign that the EV slowdown is catching up to Elon Musk.Electric vehicle shoppers are getting sick of Tesla.Well over half of the E... [+243 chars]"
    },
    - {
      - source: {
        id: null,
        name: "Biztoc.com"
      },
      author: "arstechnica.com",
```

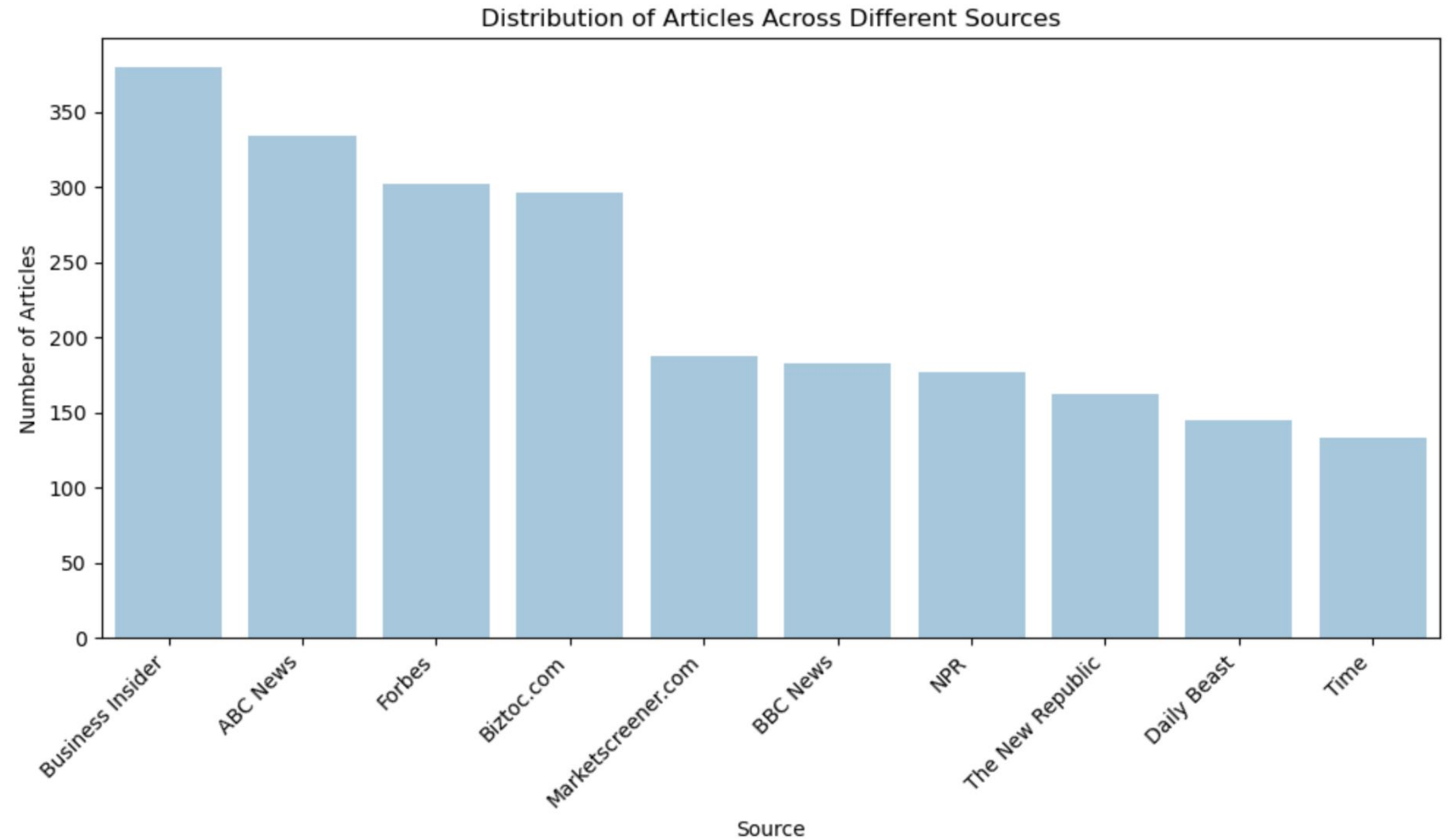
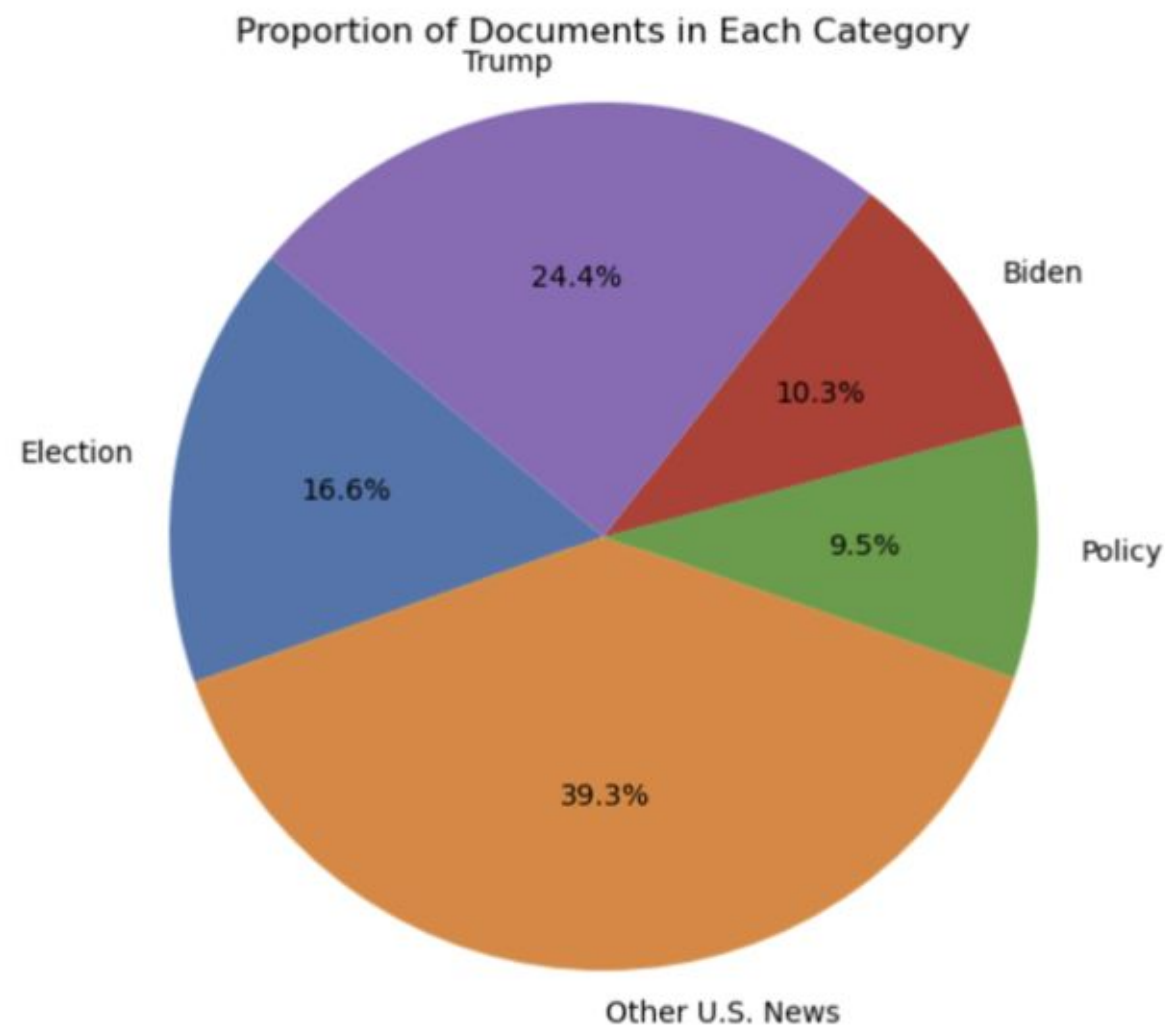
Data Processing

1. Data cleaning and categorization based on keywords in the headline, description, and content
2. Filtering criteria: Presence of predetermined keywords like '**2024 U.S. Election**', '**Primary**', '**President**', '**Biden**', '**Trump**', etc.
3. Categorization: Articles assigned to one of four categories (**Trump**, **Biden**, **Election**, **Policy/Other U.S. News**) based on content
4. Calculated Sentiment Score and Character Counts
5. Output: A JSON file named 'final_data.json' containing article details like title, description, content, category, sentiment, date, and link

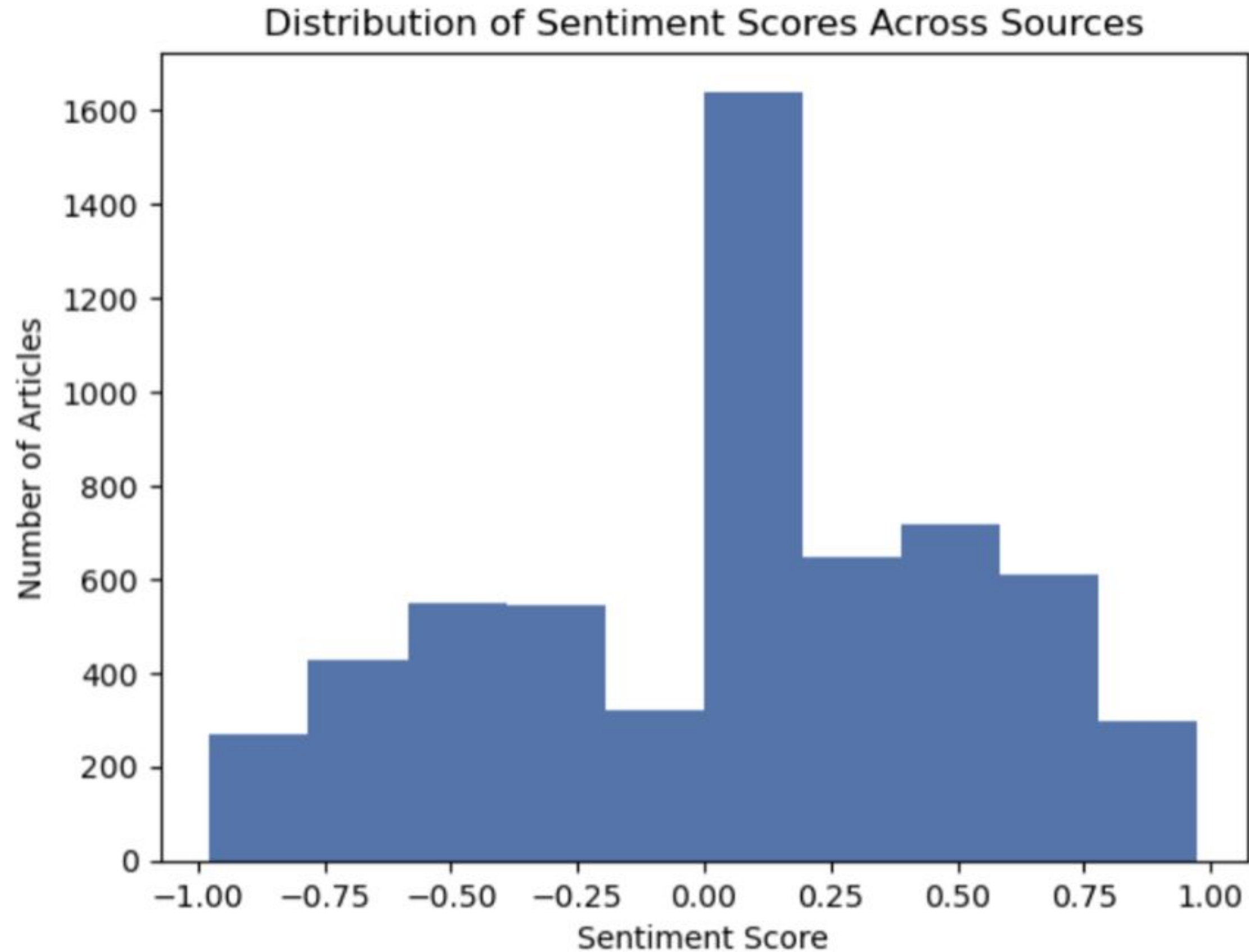
Implementation of a NewsAPI class for querying data in Mongo

```
],  
  "character_count": 4218,  
  "sentiment": -0.3182,  
  "category": "Trump"  
},
```


Exploratory Data Analysis



All News Sources

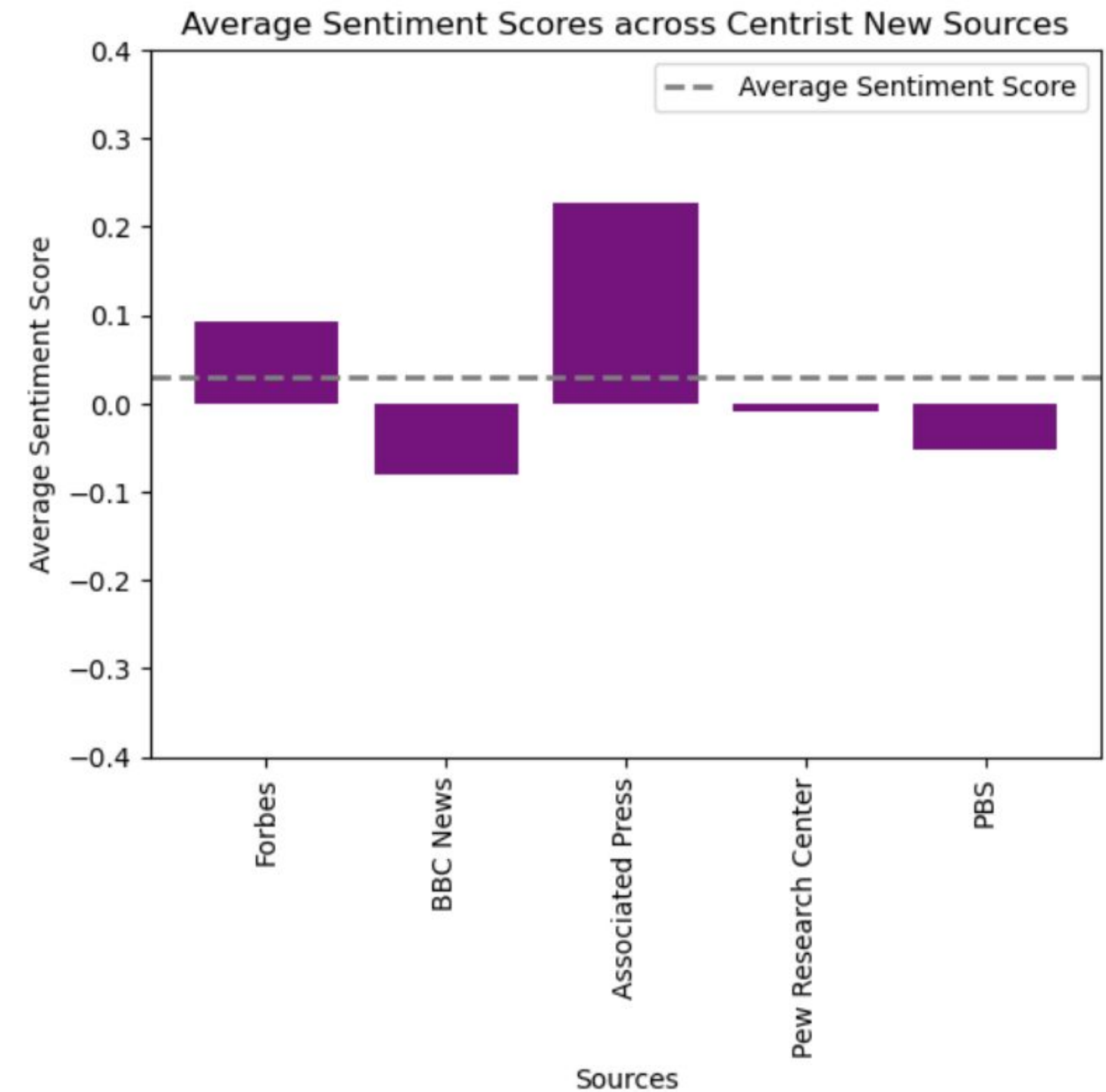
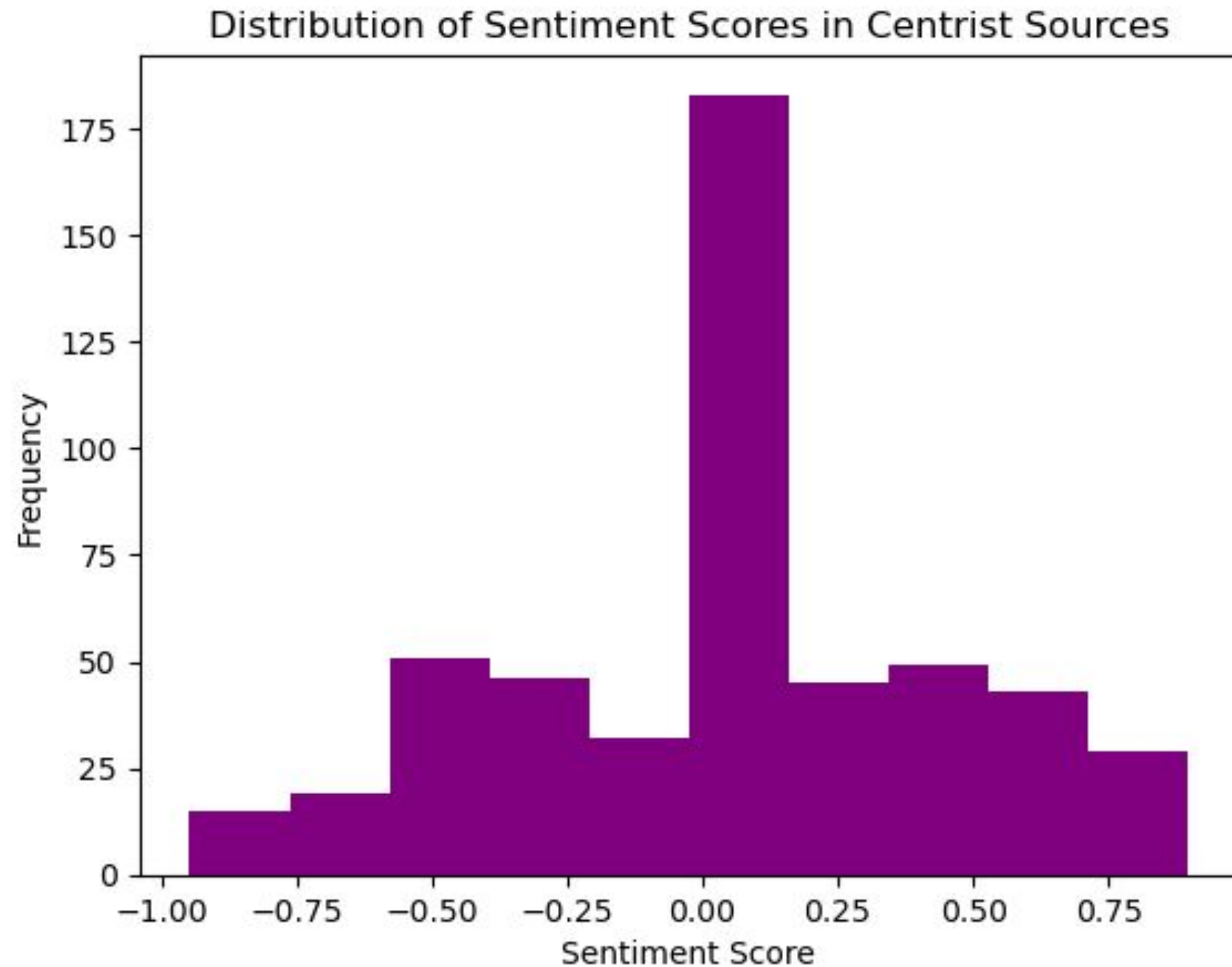


Studying Media Bias

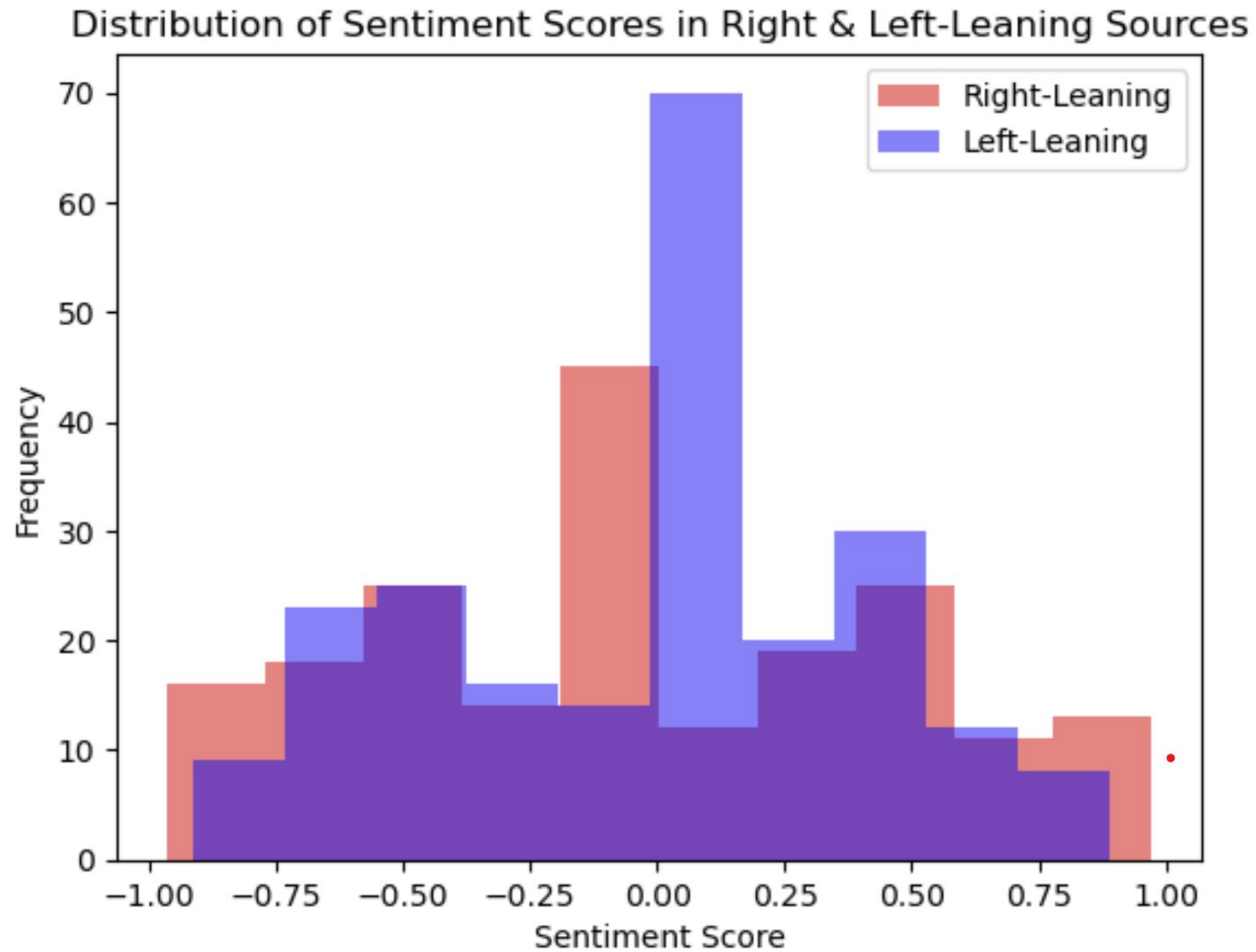
Using external sources on political bias we classified popular sources into 3 categories:

RIGHT WING	CENTRIST	LEFT-WING
● FOX News ● The Daily Caller ● Breitbart News ● The Blaze ● Washington Examiner ● Washington Free Beacon ● The Christian Post ● The American Conservative ● The Daily Caller ● The Federalist ● Americanthinker.com	● BBC News ● PBS News ● News Nation ● News Week ● The Wall Street Journal ● Reuters ● Forbes ● Associated Press ● Pew Research Center	● CNN ● HuffPost ● MSNBC ● The Washington Post ● 'Democracy Now! ● The Atlantic

Centrist Sentiment Score Distribution

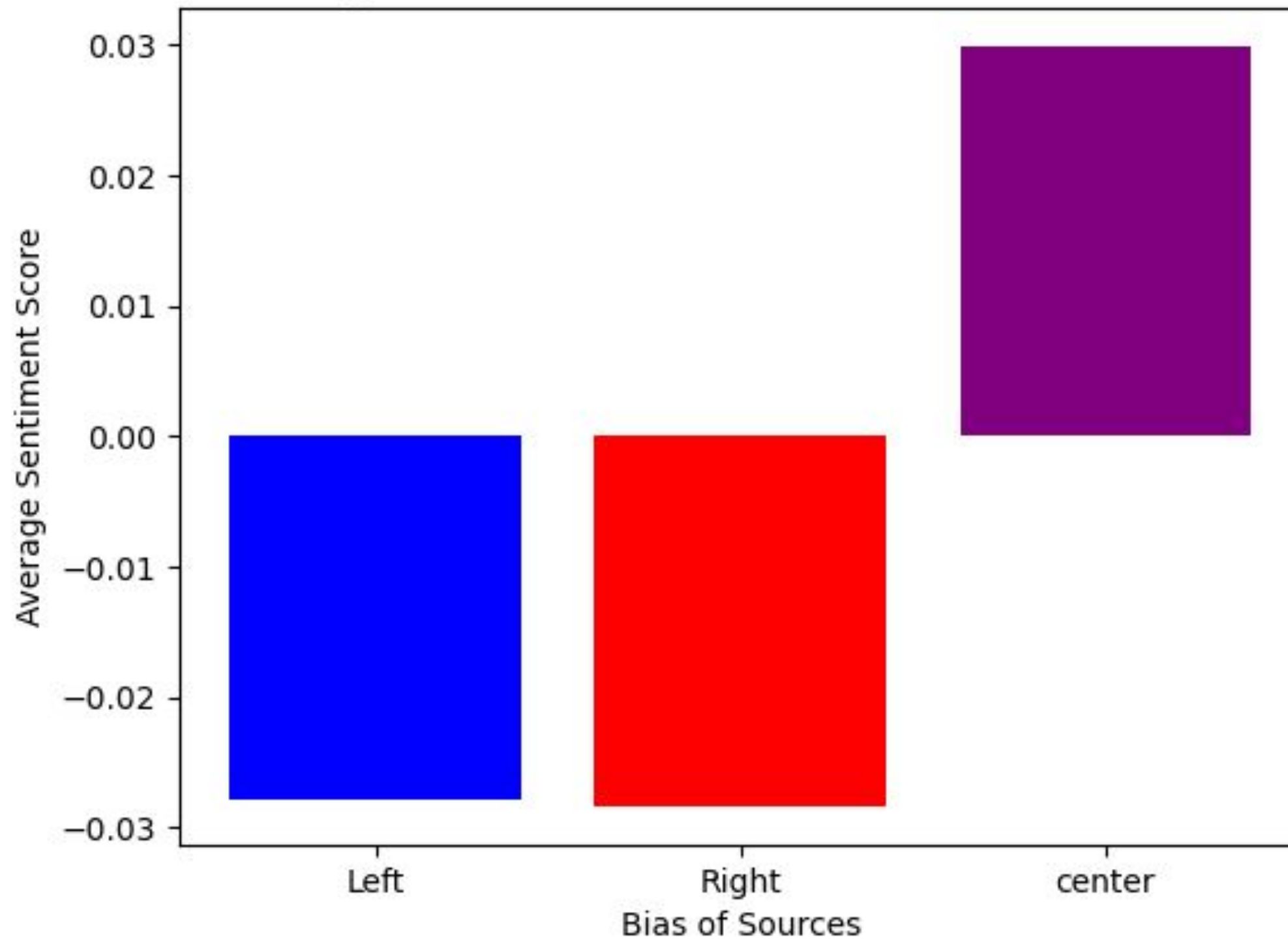


Left vs. Right Sentiment Score Distr.

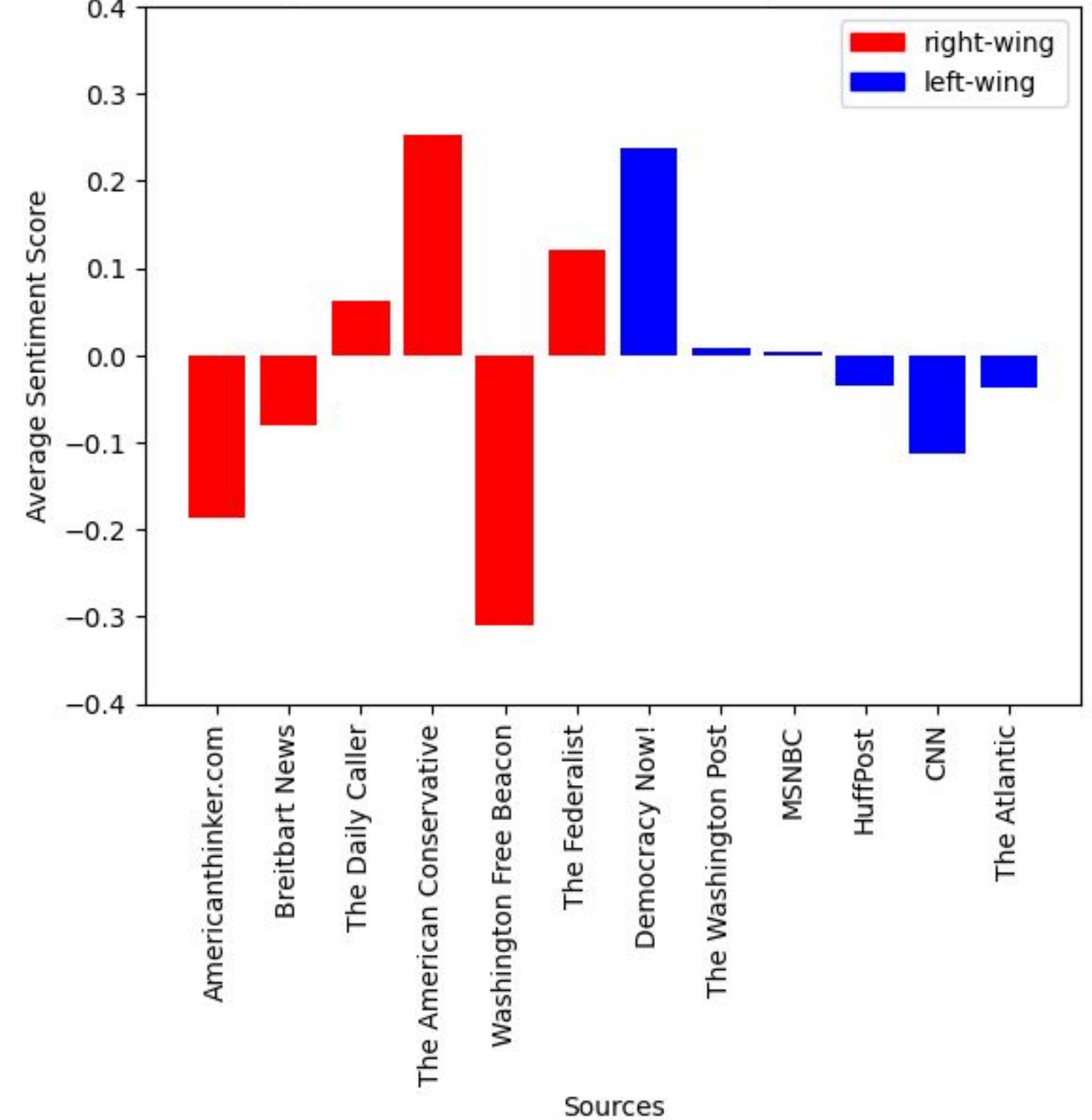


Media Bias

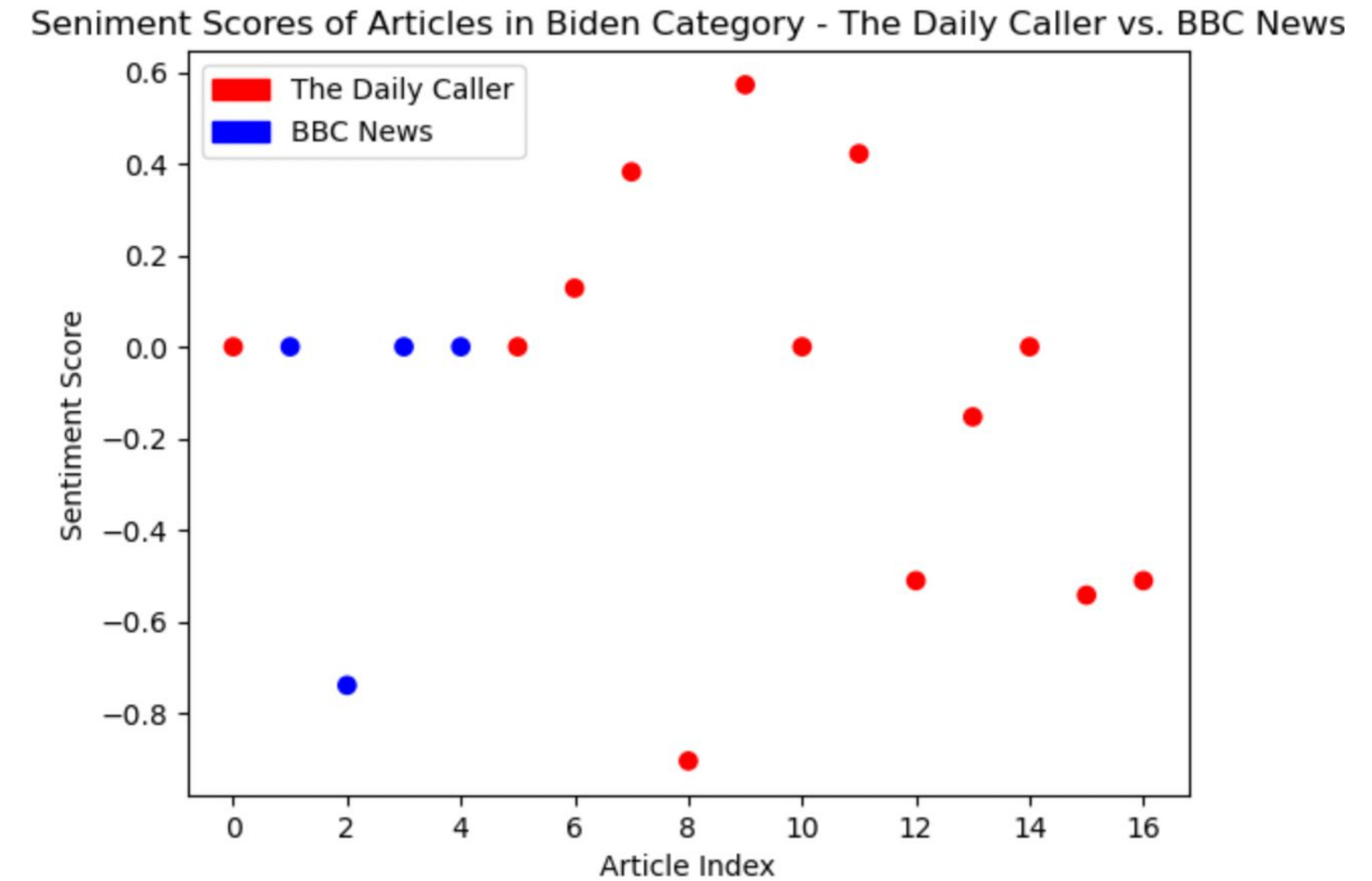
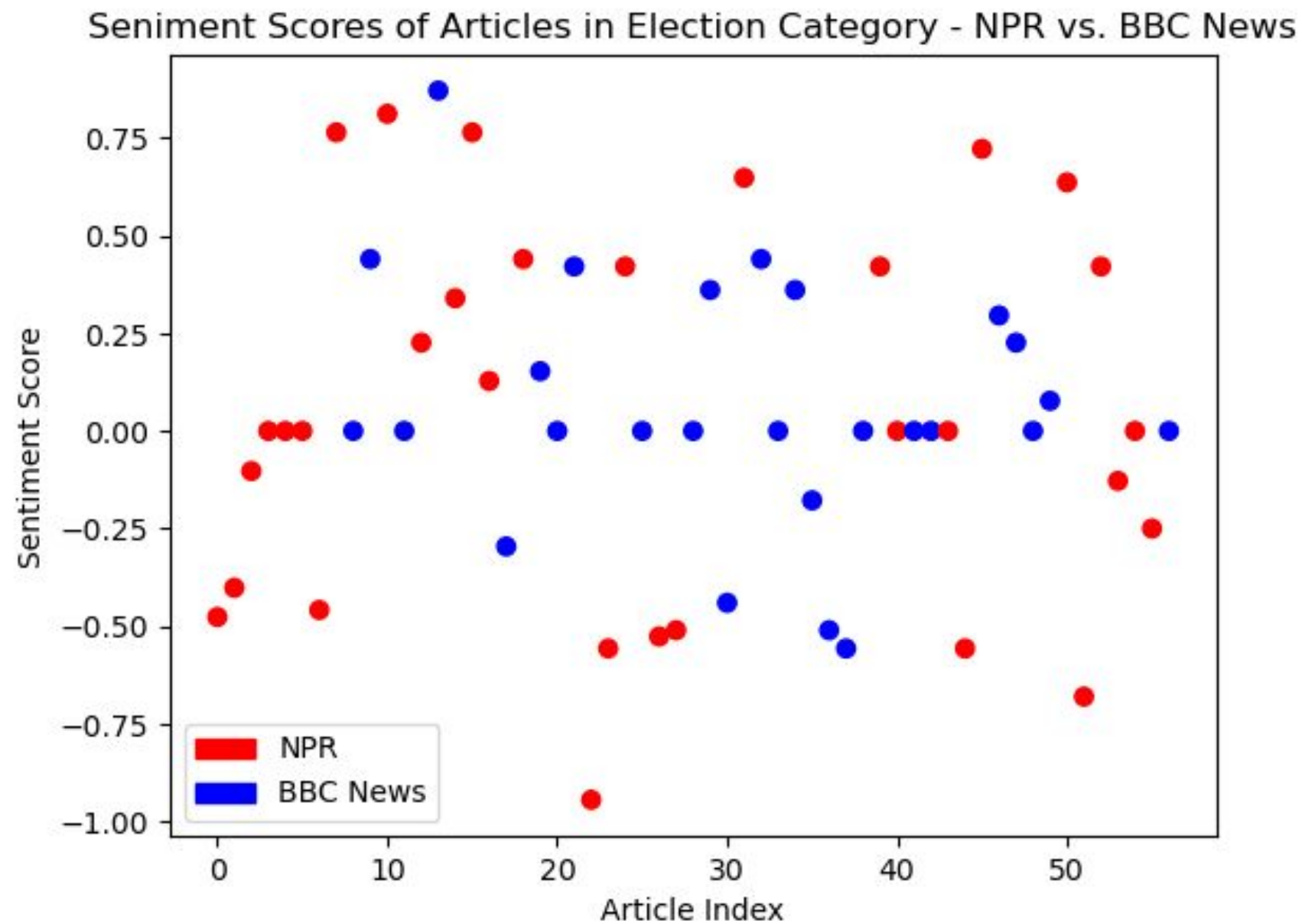
Average Sentiment Scores across Political Affiliations



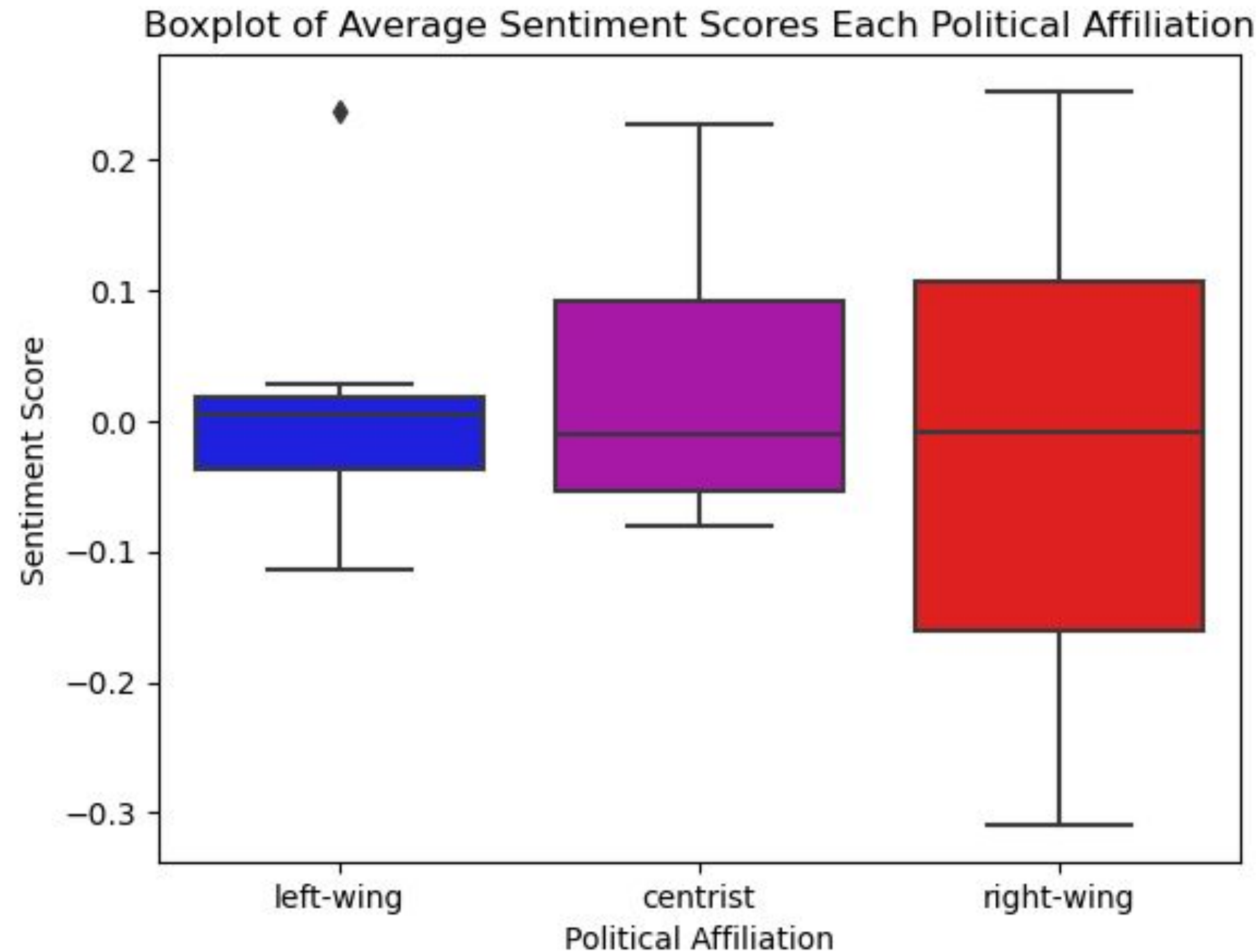
Average Sentiment Scores across Top Biased Sources



Pairwise Source Comparisons & Keywords

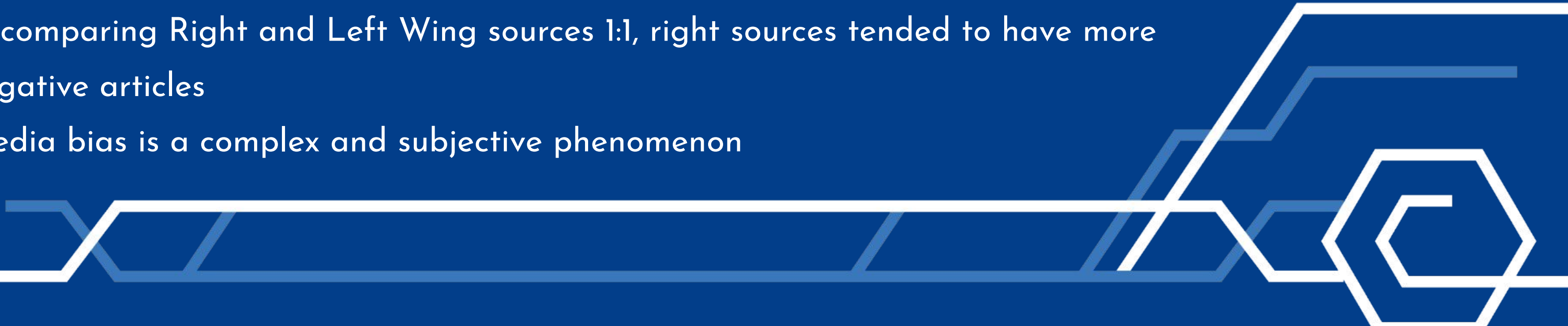


ANOVA Analysis Results



- F-statistic: 0.18
- p-value: 0.83
- Fail to reject the null hypothesis:
There is no significant difference in average sentiment scores between the news sources across political affiliations.

Conclusions

1. Overall: no significant findings on the difference of sentiment scores based on political perspective
 2. Few very skewed sources increase overall average sentiment scores across political lines
 3. Most articles have a neutral sentiment score
 4. No trends between Articles in specific Categories
 5. Found a smaller range of sentiment scores amongst Centrist sources in comparison to Right and Left Wing
 6. In comparing Right and Left Wing sources 1:1, right sources tended to have more negative articles
 7. Media bias is a complex and subjective phenomenon
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Limitations & Next Steps

Limitations:

Time range restrictions – only data from past month

Full article content not included in API data – sentiment scores not as holistic

Sentiment scoring could be an ineffective tool

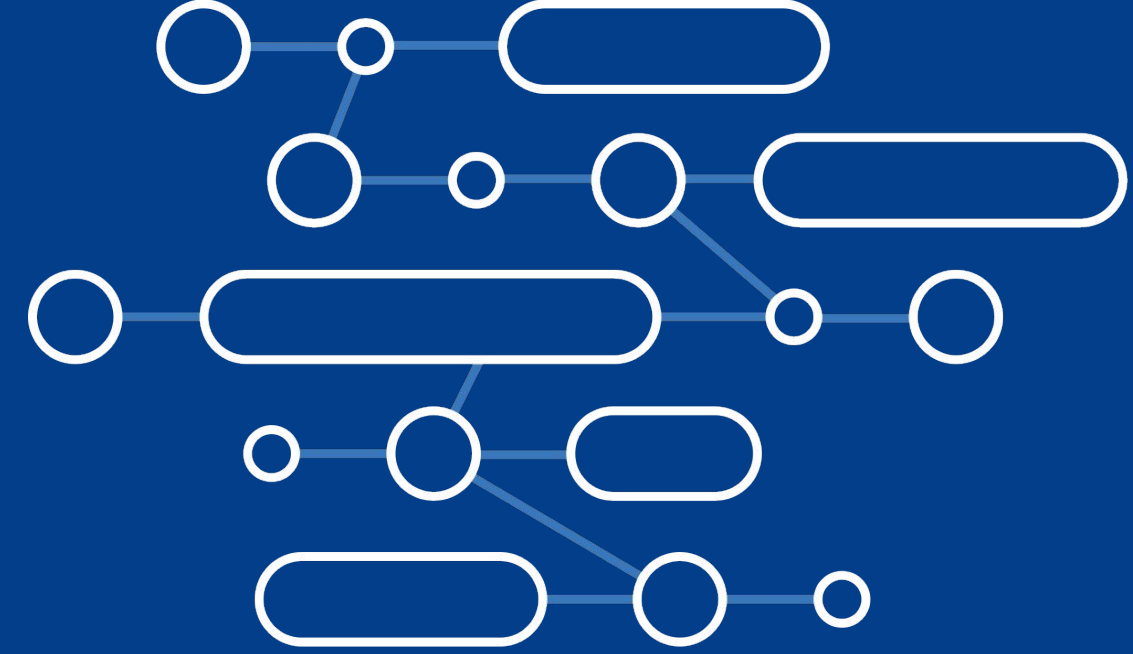
Further steps:

Analyzed quite a bit of business perspective articles – what would other perspectives look like

Scrape news sites for full article text

Run a form of text classification (split labeled dataset into training and testing)





Thank you for viewing!
Any questions?

